Supplementary Material: Benchmark Definitions, Execution Protocol, and Uncertainty

This supplement to *Black-Box Configuration Search for Generalized Disjunctive Programming* provides benchmark definitions, execution and acceptance rules, model sizes, seed-level uncertainty, and the numerical evidence underlying the algebraic bound-consistency checks. The Grid-GDP functions are adapted from the Quantum Hamiltonian Descent collection; each instance is defined by its implemented expression and domain rather than its conventional name (SECQUOIA, 2026a; Wu Group, n.d.).

S1 Coordinate domains and normalization

For each row of Table S1, both endpoints are included and the physical coordinates are

$$a_i = \ell_x + \frac{i}{30}(u_x - \ell_x), \qquad b_j = \ell_y + \frac{j}{30}(u_y - \ell_y), \qquad i, j \in \{0, \dots, 30\}. \tag{S1}$$

The raw objective f is defined below. The minimized objective is

$$F(x, y) = \frac{f(x, y) - L}{U - L}, \qquad L = \min_{i,j} f(a_i, b_j), \quad U = \max_{i,j} f(a_i, b_j). \tag{S2}$$

Thus L and U are obtained by exhaustive preprocessing of the 961 lattice points, not from the continuous optimum. All trigonometric arguments are in radians; $e = \exp(1)$ and π is the usual circular constant.

Table S1: Grid-GDP benchmark domains. All intervals include both endpoints and have equally spaced coordinates. Each row defines one independent problem with 961 configurations and 62 Boolean indicators. Historical function expressions and domains are preserved (SECQUOIA, 2026a; Wu Group, n.d.).

Function	x domain	y domain	Grid
Ackley	$[-5, 5]$	$[-5, 5]$	31×31
Alpine 1	$[0, 10]$	$[0, 10]$	31×31
Alpine 2	$[0, 1000]$	$[0, 1000]$	31×31
Bohachevsky 2	$[-100, 100]$	$[-100, 100]$	31×31
Three-hump camel	$[-5, 5]$	$[-5, 5]$	31×31
Csendes (squared variant)	$[-1, 1]$	$[-1, 1]$	31×31
Deflected corrugated spring	$[0, 10]$	$[0, 10]$	31×31
Drop-Wave	$[-5.12, 5.12]$	$[-5.12, 5.12]$	31×31
Easom	$[-10, 10]$	$[-10, 10]$	31×31
Griewank	$[-600, 600]$	$[-600, 600]$	31×31
Hölder table	$[-10, 10]$	$[-10, 10]$	31×31
Hosaki	$[0, 5]$	$[0, 6]$	31×31
Lévy	$[-10, 10]$	$[-10, 10]$	31×31
Lévy 13	$[-10, 10]$	$[-10, 10]$	31×31
Michalewicz	$[0, \pi]$	$[0, \pi]$	31×31
Rastrigin	$[-5.12, 5.12]$	$[-5.12, 5.12]$	31×31
Rosenbrock	$[-2, 2]$	$[-1, 3]$	31×31
Shubert (historical variant)	$[0, 10]$	$[0, 10]$	31×31
Styblinski–Tang	$[-5, 5]$	$[-5, 5]$	31×31
Sum of squares	$[-10, 10]$	$[-10, 10]$	31×31
Xin-She Yang 3 (historical variant)	$[-20, 20]$	$[-20, 20]$	31×31

S2 Grid-GDP objective functions

Each expression below defines a benchmark on the corresponding domain in Table S1, with all coefficients fixed. The definitions follow `gdp_models/grid.py`; differences from the historical `qhdopt_gdp.py` are noted where relevant (SECQUOIA, 2026a, n.d.b).

1. Ackley

$$f(x,y) = -20\exp\left(-0.2\sqrt{\frac{x^2+y^2}{2}}\right) - \exp\left(\frac{\cos(2\pi x)+\cos(2\pi y)}{2}\right) + 20 + e. \tag{S3}$$

2. Alpine 1

$$f(x,y) = |x\sin x + 0.1x| + |y\sin y + 0.1y|. \tag{S4}$$

3. Alpine 2

$$f(x,y) = -\sqrt{xy}\,\sin x \sin y. \tag{S5}$$

The negative sign and the domain $[0,1000]^2$ are retained.

4. Bohachevsky 2

$$f(x,y) = x^2 + 2y^2 - 0.3\cos(3\pi x)\cos(4\pi y) + 0.3. \tag{S6}$$

5. Three-hump camel

$$f(x,y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2. \tag{S7}$$

6. Csendes (squared variant)

$$f(x,y) = c(x) + c(y), \qquad c(t) = \begin{cases} t^6\left[2+\sin(1/t)\right]^2, & t \neq 0, \\ 0, & t = 0. \end{cases} \tag{S8}$$

The square applies to the entire oscillatory factor. The current GDP uses branch-fixed auxiliaries $r_t = 1/t$ for $t \neq 0$ and $r_t = 0$ for $t = 0$, evaluating $t^6[2+\sin(r_t)]^2$ without division by zero.

The archived `qhdopt_gdp.py` instead uses $c_\varepsilon(t) = t^6[2+\sin(1/(t+\varepsilon))]^2$ with $\varepsilon = 10^{-12}$ (SECQUOIA, n.d.b). This also gives zero at $t = 0$, but is not algebraically identical to Equation (S8) at nonzero coordinates. The current implementation uses the exact zero extension, with no epsilon shift; its normalized values agree with the archived grid to the stated 10^{-12} absolute verification tolerance.

7. Deflected corrugated spring

With $r = \sqrt{(x-5)^2 + (y-5)^2}$,

$$f(x,y) = 0.1r^2 - \cos(5r). \tag{S9}$$

8. Drop-Wave

$$f(x,y) = -\frac{1+\cos\left(12\sqrt{x^2+y^2}\right)}{0.5(x^2+y^2)+2}. \tag{S10}$$

9. Easom

$$f(x,y) = -\cos x \cos y\, \exp\left(-[(x-\pi)^2 + (y-\pi)^2]\right). \tag{S11}$$

10. Griewank

$$f(x,y) = \frac{x^2+y^2}{4000} - \cos x \cos\left(\frac{y}{\sqrt{2}}\right) + 1. \tag{S12}$$

11. Hölder table

$$f(x,y) = -\left|\sin x \cos y \, \exp\left(\left|1 - \frac{\sqrt{x^2+y^2}}{\pi}\right|\right)\right|. \tag{S13}$$

12. Hosaki

$$f(x,y) = \left(1 - 8x + 7x^2 - \frac{7}{3}x^3 + \frac{1}{4}x^4\right) y^2 \exp(-y). \tag{S14}$$

13. Lévy

Define $w_x = 1 + (x-1)/4$ and $w_y = 1 + (y-1)/4$. Then

$$\begin{aligned} f(x,y) = \; & \sin^2(\pi w_x) \\ & + (w_x - 1)^2\left[1 + 10\sin^2(\pi w_x + 1)\right] \\ & + (w_y - 1)^2\left[1 + \sin^2(2\pi w_y)\right]. \end{aligned} \tag{S15}$$

The middle sine argument is $\pi w_x + 1$.

14. Lévy 13

$$\begin{aligned} f(x,y) = \; & \sin^2(3\pi x) + (x-1)^2\left[1 + \sin^2(3\pi y)\right] \\ & + (y-1)^2\left[1 + \sin^2(2\pi y)\right]. \end{aligned} \tag{S16}$$

15. Michalewicz

$$f(x,y) = -\sin x \, \sin^{20}\left(\frac{x^2}{\pi}\right) - \sin y \, \sin^{20}\left(\frac{2y^2}{\pi}\right). \tag{S17}$$

The fixed exponent is 20 (equivalently $2m$ with $m = 10$).

16. Rastrigin

$$f(x,y) = x^2 - 10\cos(2\pi x) + y^2 - 10\cos(2\pi y) + 20. \tag{S18}$$

17. Rosenbrock

$$f(x,y) = (1-x)^2 + 100(y - x^2)^2. \tag{S19}$$

18. Shubert (historical variant)

$$\begin{aligned} f(x,y) = \; & \left[\cos(2x+1) + 2\cos(3x+2) + 3\cos(4x+3)\right] \\ & \times \left[\cos(2y+1) + \cos(y+2)\right]. \end{aligned} \tag{S20}$$

This is the implemented three-term by two-term product. In particular, the final term is $\cos(y+2)$ with coefficient one.

19. Styblinski–Tang

$$f(x,y) = \frac{1}{78}\,\frac{1}{2}\left(x^4 - 16x^2 + 5x + y^4 - 16y^2 + 5y\right). \tag{S21}$$

The historical factor $1/78$ is retained in the raw objective, even though positive scaling cancels in the normalization of Equation (S2).

20. Sum of squares

$$f(x, y) = x^2 + 2y^2. \tag{S22}$$

21. Xin-She Yang 3 (historical variant)

$$f(x, y) = \exp\left(-\left[\left(\frac{x}{15}\right)^6 + y^6\right]\right) + 2\exp\bigl(-(x^2 + y^2)\bigr)\cos^2 x \cos^2 y. \tag{S23}$$

The exponential–cosine term has a positive coefficient +2. Only x is divided by 15 in the first exponential; y^6 is unscaled. Floating-point exponential underflow can produce stored zero values; these do not change the analytical expression.

S3 Source verification

Exhaustive evaluation of the implemented Grid-GDP objectives at all $21 \times 961 = 20{,}181$ lattice points agrees with the archived `benchmark_grids/*.csv` normalized values within absolute tolerance 10^{-12}. The target sets at $F \leq 10^{-5}$ also agree.

S4 Executable protocol and implementation settings

Python and NumPy use the job seed, and `RBFOpt` also receives `rand_seed`. `BAM` exposes no separate native random-seed option: its process wrapper seeds Python/NumPy, while the Grid seed identifies the common initial design. `FMQA-SA` uses three NumPy `SeedSequence(seed)` streams for FM initialization, annealing, and uniform unseen-point fallback; each training or annealing invocation draws a seed from its stream. `PyTorch` uses deterministic CPU operations and one thread.

Repeated callbacks reuse the stored response and point cost. Unaccepted process evaluations return 10^6 to `D-SDA` and `RBFOpt`.

Grid `D-SDA` ranks the initial points and attempts successive descents within the evaluation budget. After each completed descent, it applies the common target criterion to the best observed objective before trying the next ranked start, unless the budget is already exhausted. It enumerates the full L_∞ neighborhood as the lexicographic Cartesian product of $(-1, 0, 1)$, excluding the zero direction and out-of-bounds points; ties select the first point, and improvement must exceed 10^{-5}.

Primary `RBFOpt` 4.3.0 uses MSRSM, `rbf=auto`, the default genetic candidate-search option `global_search_method=genetic`, integer variables, and `do_init_strategy=False`. The recorded defaults are `ga_base_population_size=400` and `ga_num_generations=20`. `RBFOpt` 4.3.0 uses an effective population of $400 + \lfloor 20n/5 \rfloor$, where n is the dimension passed to the genetic-search routine, giving 408 and 412 for two and three search variables. Standalone `Bonmin` 1.8.9 (Bonami et al., 2008) solves surrogate MINLPs, and the configured `IPOPT` 3.14.19 executable supports surrogate NLPs (Wächter and Biegler, 2006). `BAM` 5.0.0 receives `x0=None`, all-integer variables, the three evaluated starts, and `max_evals`=B-3, with other options at package defaults; its local models use `ALAMO`/`BARON`.

`FMQA-SA` stops at B or its 1,800 s method limit, checked between operations and through training/sampling callbacks, without reference-target stopping. Each FM uses identity activation, zero bias, and normal initial weights with mean zero and standard deviation 0.01. Training uses Adam (Kingma and Ba, 2015), 1,000 full-batch epochs, mean squared error, and learning rate 0.01. Coordinate ξ_j uses $\lfloor \log_2 u_j \rfloor + 1$ unsigned bits, where $u_j > 0$ is its upper bound; QUBO coefficients are divided by the largest absolute entry of the quadratic matrix. D-Wave's `SimulatedAnnealingSampler` in `dwave-samplers` 1.6.0 uses 100 reads, 1,000 sweeps, and a geometric inverse-temperature schedule (D-Wave Systems, n.d.). All process workers and the Grid `FMQA-SA` worker have a 2,100 s watchdog. The Grid `RBFOpt`/`BAM` trace watchdogs are disabled (`max_run_seconds=0`); archived Grid `D-SDA` uses budget, target, or local-search termination without an additional method-clock limit.

S4.1 Versions and independently timed process evaluations

The recorded environments include Python 3.11.16, `RBFOpt` 4.3.0 (Costa and Nannicini, 2018; Nannicini, 2021), `BAM` 5.0.0 (Ma et al., 2023), `Pyomo` 6.10.0 (Bynum et al., 2021), and `SCIP` 9.2.4 (Bolusani et al., 2024). The annealer is the classical CPU implementation `dwave.samplers.SimulatedAnnealingSampler` (D-Wave Systems, n.d.). Executable banners identify `Bonmin` 1.8.9 with `Cbc` 2.10.12 and `IPOPT` 3.14.19, and standalone `IPOPT` 3.14.19, on Linux x86_64 with ASL 20231111.

The `bam-solver` 5.0.0 engine identifies version 2026.9.10, build LNX-int-64, timestamp 2026-09-10 17:59:06. `BAM`'s internal `ALAMO` and `BARON` versions are unspecified. The algebraic comparisons use `SCIP` 9.2.4 through `GAMS` 52.2.0 for MINLP reformulations and both `GDPOpt` subsolvers. Grid preprocessing fixes all original variables, and the recorded `GDPOpt` Grid runs make no NLP solver calls. The algebraic process models retain their archived initializations, including constructor defaults for `CSTR30`; the updated DFO evaluation table is identified separately in Section S8.

The fixed-configuration process evaluations use `IPOPT` 3.14.19 (Wächter and Biegler, 2006) with `MUMPS` 5.8.1 (Amestoy et al., 2001), solver tolerance 10^{-8} and a 60 s solver limit. The acceptance rule in `gdp_models/evaluation.py` requires all of the following: a finite objective; termination in {`optimal`, `locallyOptimal`, `globallyOptimal`}; solver status other than `error` or `aborted`; interface status `SOLVESTAT`=1 and `MODELSTAT`$\in \{1, 2\}$; and a finite maximum primal violation no greater than 10^{-5}. The 10^{-8} solver tolerance is distinct from this acceptance threshold. Time-limit, iteration-limit, evaluation-limit, and error outcomes are not accepted, even if a finite feasible candidate is available.

The primal check takes the largest positive absolute constraint-bound or variable-bound violation, with zero for satisfied bounds and no row-wise normalization. Constraint rows are collected after fixing the disjunctions and before preprocessing, so rows subsequently deactivated by presolve are also checked. Bounds of initialized active variables are checked on the prepared NLP. An undefined or non-finite constraint body, a non-finite initialized variable, or a non-finite objective precludes acceptance. Uninitialized unused variables are skipped; the objective and all retained constraint expressions are still evaluated. Preprocessing rejection or worker failure supplies no accepted objective.

`CSTR5`–`CSTR30` start from constructor defaults and then apply `cstr_box_ipopt.py:initialize_feasible` after preprocessing. This sets mass-balance-consistent values with zero recycle and checks the initial primal violation at 10^{-8} without changing constraints or bounds. The `CSTR30` table retains one interface failure, which none of the 50 searches visited. `Column` and `Small Batch` load `gdp/dsda/column_17_initialization.json` and `gdp/dsda/small_batch_initialization.json`, respectively, from the recorded `dsda-gdp` source revision, using its `gdp/dsda/model_serializer.py`. `Dynamic` clones the unsolved discretized template; it has no saved initialization vector. These are cold starts, with no transfer of an NLP solution between configurations. Objectives, acceptance status, and NLP point costs are linked to the same recorded attempt by model, `CSTR` size, and coordinate, without successful-retry selection or cross-size time substitution. The `CSTR30` point costs include initialization and artifact recording. Separately measured structural-rejection costs cover 435 `CSTR30` and 35 `Dynamic` coordinates without invoking an NLP. Each stored cost is reused on revisits, so reconstructed costs do not capture repeat-measurement timing variability; `Dynamic` template preparation is added once per run.

Process lookup-call times cover coordinate processing and in-memory lookup, excluding callback-file writing; the CSV is loaded before the search clock starts. Recorded search time includes callback logging and termination checks, but excludes final incumbent extraction, summary writing, and export.

S4.2 Observed proposal and initialization behavior

Table S2 pools `FMQA-SA` proposals across tested budgets and ten seeds per condition, including all six `CSTR` sizes. Counts exclude the shared initialization and include terminal-callback proposals; rates are pooled fallback/proposal ratios. Uniform fallback supplies most process proposals, so the results characterize the combined learned-proposal and fallback procedure. The added uniform control compares the complete `FMQA-SA` procedure with sampling under the shared protocol; it does not isolate the individual causal contributions of fitting, annealing, and fallback.

Table S2: Recorded `FMQA-SA` uniform fallback by benchmark family. Counts include proposals for terminal callbacks.

Family	Runs	Proposals	Fallbacks	Rate
Grid	1,050	51,450	9,522	18.51%
`CSTR`	60	1,350	369	27.33%
`Column`	10	100	86	86.00%
`Dynamic`	60	1,460	1,063	72.81%
`Small Batch`	10	70	58	82.86%
Process total	140	2,980	1,576	52.89%

S4.3 Terminal-callback solutions and final summaries

Process success is defined by the wrapper's best accepted configuration, including callback B, rather than by the native optimizer's returned incumbent. The final `runs.csv` summary stores the coordinates and objective of that configuration.

S5 Process model scale, units, and discrete decisions

Counts below describe the unfixed original GDP before reformulation or presolve, including all active disjunct branches, not the smaller NLP for a particular selected configuration. Fixed continuous variables remain in the continuous-variable count; the numerical suffix in `CSTR5`–`CSTR30` specifies the number of potential reactor positions in the superstructure. `Dynamic` is discretized with 30 finite elements per stage and three Radau points. Binary counts refer to the indicator variables already constructed by the model, before Boolean-to-binary conversion. All nine models have zero additional nonbinary integer variables; the process GDP formulations appear in Appendices A.1–A.3 of Ovalle et al. (2025), with external-variable mappings in Appendices B.1, B.2, and B.4, and the `Dynamic` formulation appears in Section IV-C of Peng et al. (2024).

Table S3: Original process-model sizes. n_c: continuous variables; n_b: binary variables; m_a: algebraic constraints; m_l: logical constraints; n_d: disjunctions.

Model	n_c	n_b	m_a	m_l	n_d
`CSTR5`	56	20	100	12	10
`CSTR10`	106	40	195	22	20
`CSTR15`	156	60	290	32	30
`CSTR20`	206	80	385	42	40
`CSTR25`	256	100	480	52	50
`CSTR30`	306	120	575	62	60
`Column`	446	30	637	16	15
`Small Batch`	19	18	34	3	9
`Dynamic`	1647	27	3276	31	9

For `CSTR`, the cutoff/feed and recycle choices determine reactor/bypass and recycle disjuncts; the objective is total reactor volume in litres (Liñán et al., 2020). `Column` selects reflux/boilup positions and the implied tray configuration (Ghouse et al., 2018). Its implemented objective is $10^3(Q_c + Q_b) + 10^3 n_{\text{active}}$, a weighted cost proxy labelled in dollars by the source model; it is not a newly estimated annualized economic cost. `Small Batch` selects the parallel-unit counts for three stages, fixing the discrete unit-number alternatives; its source objective is investment cost in dollars (Kocis and Grossmann, 1988). No currency-year conversion is applied. `Dynamic`'s two switching-stage choices fix each stage's mode and leave the continuous state trajectories to the discretized NLP (Peng et al., 2024). Its objective is $-\int_0^9 x(t)^2\,dt$, so minimization favors a larger integrated squared state. State and time are abstract benchmark quantities without a physical unit calibration in the implementation. The example adds mode sequencing and dynamic differential constraints to the static equipment/configuration cases; it is not a plant-specific economic metric.

`Column` uses the logical domain $\{8,\dots,15\} \times \{1,\dots,8\}$ (64 points). These bounds encode reflux at or above the feed tray and boilup at or below it. The remaining algebraic constraints are checked during configuration evaluation. All search methods and uniform sampling use this domain; the algebraic references retain the original GDP formulation.

S6 Grid inventory and target-set sizes

All DFO methods receive the integer pair $(i, j) \in \{0,\dots,30\}^2$. The one-based external-variable convention in the main paper corresponds to $(\xi_1, \xi_2) = (i + 1, j + 1)$; this is a relabelling, not an offset applied by the Grid optimizers. Physical coordinates follow Equation (S1). `FMQA-SA` uses five unsigned bits per Grid coordinate; decoded coordinate values equal to 31 are rejected as out of bounds. The normalized tolerance 10^{-5} equals $10^{-5}(U - L)$ in raw objective units. Target-set sizes therefore vary across functions even though the normalized tolerance is shared.

The inventory in Table S4 is calculated from all 961 implemented raw samples. Constants are rounded for display; full precision, physical coordinates, and complete minimum/target lists are in `grid_inventory.json`. $n_{\min}^{\mathrm{fp}}$ counts exact ties at the smallest floating-point raw sample, and n_{target} counts normalized samples at most 10^{-5}. Floating arithmetic may split analytic ties: Rosenbrock has two exact minimizers, although only one sample equals the floating minimum bit for bit. For Xin-She Yang 3 (historical variant), underflow gives 12 exact stored minima, and 870 grid points meet the target. Neither exact stored zeros nor target membership imply that the corresponding continuous analytical function is zero there.

Table S4: Enumerated grid normalization constants and target-set sizes.

Function	L	U	$n_{\min}^{\mathrm{fp}}$	n_{target}
Ackley	4.44089e-16	14.2469	1	1
Alpine 1	0	17.4297	1	1
Alpine 2	-896.050	789.540	1	1
Bohachevsky 2	0	30000	1	1
Three-hump camel	0	2047.92	1	1
Csendes (squared variant)	0	16.1479	1	36
Deflected corrugated spring	-1	5.69827	1	1
Drop-Wave	-1	-1.31451e-7	1	1
Easom	-0.895368	0.00846156	1	1
Griewank	0	180.012	1	1
Hölder table	-18.1380	-0	2	4
Hosaki	-2.34581	0.541341	1	1
Lévy	0.114062	95.3828	1	2
Lévy 13	0.305556	307.861	1	4
Michalewicz	-1.80107	-0	1	1
Rastrigin	0	77.8494	1	1
Rosenbrock	0.00641975	2509	1	2
Shubert (historical variant)	-11.0378	10.7203	1	1
Styblinski–Tang	-1	3.20513	1	1
Sum of squares	0	300	1	1
Xin-She Yang 3 (historical variant)	0	3	12	870

S7 Success-rate uncertainty and completion costs

For each problem, method, and budget, let h be the number of target hits among $n = 10$ seeds, retaining execution failures as non-successes.

We report two-sided 95% Clopper–Pearson intervals for the binomial success probability (Clopper and

Pearson, 1934):

$$p_L = \begin{cases} 0, & h = 0, \\ B^{-1}_{0.025}(h, n-h+1), & h > 0, \end{cases} \qquad p_U = \begin{cases} 1, & h = n, \\ B^{-1}_{0.975}(h+1, n-h), & h < n, \end{cases}$$

where B^{-1} denotes the beta quantile. Coverage is conditional on treating the seed-level outcomes as independent Bernoulli trials under the fixed protocol. The intervals are pointwise, not simultaneous across comparisons.

Ten observed successes give $[0.692, 1]$; zero give $[0, 0.308]$. Thus neither empirical endpoint establishes a population probability of one or zero. Table S5 reports completion costs for all 14 process conditions.

Table S6 reports function-level results at the largest Grid budget, $B = 100$. Complete seed-level outcomes for all Grid budgets and Process conditions are available in the publication analysis linked in Section S8. Cells report $h\,[p_L, p_U]$, with limits rounded to two decimals; displayed endpoints may be rounded interior bounds. Seed indices remain paired across methods within each problem/budget, but intervals are calculated separately for each group, not by pooling the heterogeneous 21-function suite.

Table S5: Median reconstructed full-run completion costs for all ten paired seeds per condition, including target misses, in seconds.

Case	B	D-SDA	RBFOpt	BAM	FMQA-SA	Uniform
Column	13	35.44	54.60	27.50	41.82	21.69
CSTR5	8	10.67	10.72	10.67	13.92	9.44
CSTR10	15	18.95	21.79	18.54	27.07	17.75
CSTR15	22	29.93	37.59	27.67	42.82	27.84
CSTR20	29	40.68	45.03	39.37	59.10	40.12
CSTR25	36	51.69	62.29	50.47	74.70	51.17
CSTR30	43	53.12	101.96	115.41	101.19	65.20
Small Batch	10	3.84	5.12	4.23	9.09	3.49
Dynamic	12	12.23	12.29	9.95	14.47	7.80
Dynamic	15	15.85	16.10	11.53	18.39	8.71
Dynamic	20	21.30	20.54	14.07	25.23	12.27
Dynamic	30	26.62	34.19	21.91	36.24	19.38
Dynamic	37	26.62	42.42	24.99	46.05	24.38
Dynamic	50	26.62	65.14	33.98	61.31	33.12

The table reports completion costs, not first-hit times. Reconstructed search-completion costs exclude the final wrapper result-extraction and summary-writing pass. Pointwise success-rate intervals are defined in Supplementary Section S7. Every group has complete measured point-cost coverage, with no zero filling or cross-size substitution.

Table S6: Grid target hits and pointwise 95% success-probability intervals at $B = 100$. Each cell is $h\,[p_L, p_U]$ from ten seeds.

Function	D-SDA	RBFOpt	BAM	FMQA-SA	Uniform	BAM-prefix control
Ackley	0 [0.00, 0.31]	10 [0.69, 1.00]	10 [0.69, 1.00]	8 [0.44, 0.97]	1 [0.00, 0.45]	10 [0.69, 1.00]
Alpine1	1 [0.00, 0.45]	10 [0.69, 1.00]	10 [0.69, 1.00]	5 [0.19, 0.81]	1 [0.00, 0.45]	10 [0.69, 1.00]
Alpine2	1 [0.00, 0.45]	3 [0.07, 0.65]	1 [0.00, 0.45]	2 [0.03, 0.56]	1 [0.00, 0.45]	2 [0.03, 0.56]
Bohachevsky2	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	9 [0.55, 1.00]	1 [0.00, 0.45]	10 [0.69, 1.00]
Camel3	9 [0.55, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	2 [0.03, 0.56]	10 [0.69, 1.00]
Csendes	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]
Deflected corrugated spring	0 [0.00, 0.31]	1 [0.00, 0.45]	10 [0.69, 1.00]	3 [0.07, 0.65]	3 [0.07, 0.65]	10 [0.69, 1.00]
Dropwave	0 [0.00, 0.31]	2 [0.03, 0.56]	10 [0.69, 1.00]	4 [0.12, 0.74]	0 [0.00, 0.31]	10 [0.69, 1.00]
Easom	1 [0.00, 0.45]	7 [0.35, 0.93]	0 [0.00, 0.31]	3 [0.07, 0.65]	0 [0.00, 0.31]	1 [0.00, 0.45]
Griewank	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	2 [0.03, 0.56]	10 [0.69, 1.00]
Holder Table	1 [0.00, 0.45]	10 [0.69, 1.00]	6 [0.26, 0.88]	8 [0.44, 0.97]	4 [0.12, 0.74]	1 [0.00, 0.45]
Hosaki	10 [0.69, 1.00]	10 [0.69, 1.00]	3 [0.07, 0.65]	6 [0.26, 0.88]	0 [0.00, 0.31]	3 [0.07, 0.65]
Levy	2 [0.03, 0.56]	10 [0.69, 1.00]	5 [0.19, 0.81]	4 [0.12, 0.74]	0 [0.00, 0.31]	2 [0.03, 0.56]
Levy13	8 [0.44, 0.97]	10 [0.69, 1.00]	9 [0.55, 1.00]	9 [0.55, 1.00]	6 [0.26, 0.88]	2 [0.03, 0.56]
Michalewicz	6 [0.26, 0.88]	10 [0.69, 1.00]	2 [0.03, 0.56]	7 [0.35, 0.93]	0 [0.00, 0.31]	2 [0.03, 0.56]
Rastrigin	0 [0.00, 0.31]	4 [0.12, 0.74]	10 [0.69, 1.00]	5 [0.19, 0.81]	1 [0.00, 0.45]	10 [0.69, 1.00]
Rosenbrock	3 [0.07, 0.65]	10 [0.69, 1.00]	0 [0.00, 0.31]	6 [0.26, 0.88]	2 [0.03, 0.56]	1 [0.00, 0.45]
Shubert	1 [0.00, 0.45]	4 [0.12, 0.74]	1 [0.00, 0.45]	1 [0.00, 0.45]	1 [0.00, 0.45]	0 [0.00, 0.31]
Styblinski Tang	8 [0.44, 0.97]	10 [0.69, 1.00]	10 [0.69, 1.00]	7 [0.35, 0.93]	0 [0.00, 0.31]	0 [0.00, 0.31]
Sum Of Squares	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	9 [0.55, 1.00]	0 [0.00, 0.31]	10 [0.69, 1.00]
Xin She Yang3	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]	10 [0.69, 1.00]

S8 Data sources and reproducibility

The baseline publication analysis provides the original source records and reproduction instructions (SECQUOIA, 2026d). The CSTR5–25 RBFOpt results use the finite-penalty records. The CSTR30 results use the updated evaluation table in the initialization follow-up. The current D-SDA results, figures, and tables are provided by the tolerance-update publication analysis. The resulting 700-run process comparison contains 2,024 distinct visited configurations and 16,041 callbacks. The Column logical-domain records are provided by the publication analysis. The SCIP algebraic records are pinned separately: Grid Big-M/Hull, Grid GDPOpt, and all 45 process comparisons. The Grid GDPOpt summary retains each original non-error run and replaces only the 46 execution-error entries with their recovery runs. All first attempts remain archived; the retained time-limit misses and differing rerun sets prevent treating the mixed counts as a method ranking. The three presolve-disabled LOA diagnostic runs are excluded from these performance counts.

S9 Feasibility and bound consistency of the algebraic results

The bound-consistency check covers all 45 selected algebraic runs in manuscript Table 5. Table S7 lists the 19 flagged comparisons, and Table S8 gives the best saved feasible solution for each model in the current SCIP runs. The linked process analysis provides unrounded objectives, bounds, residuals, archive paths, and hashes.

For each minimization model, let f_w be the recorded objective of its best independently checked saved feasible solution, and let L denote a reported lower bound. We apply the same rule to all five methods:

$$\tau_w = 10^{-5}\max\{1, |f_w|\}, \qquad \Delta = L - f_w, \qquad \rho = \Delta/\tau_w, \qquad \dagger \iff \rho > 1. \tag{S24}$$

Missing lower bounds remain missing, not zero.

Saved vectors are loaded into the original GDP to reevaluate objectives and check active constraints, variable bounds, integrality, disjunctions, and logical validity without optimization. Violations are unscaled

original-equation residuals, not reformulation residuals. All 43 saved feasible vectors satisfy the 10^{-5} threshold; the largest violation is 9.99×10^{-7}. The two Big-M runs without accepted feasible solutions remain in the 45-run denominator. This numerical check establishes neither exact feasibility nor global optimality.

The uniform rule flags six `CSTR` GLOA runs, `Dynamic` Hull, RIC and LOA for `CSTR5`–`CSTR25`, and Hull for `CSTR5` and `CSTR10`. The last 12 discrepancies are only 1.09–4.30 times the tolerance and are sensitive to numerical feasibility conventions. These conflicting bounds and `optimal` reports provide no independent global certificate or intrinsic method ranking; their cause in GDPopt or its subsolvers remains unresolved.

Table S7: The 19 flagged bound comparisons among the 45 `SCIP` runs in manuscript Table 5. Bounds, violations, and ratios are rounded for display; flags use the unrounded $\rho > 1$ test. C identifies the comparison solution in Table S8. The flags are numerical-consistency categories, not equally strong demonstrations of invalid mathematical bounds.

ID	Model	Method	Reported L	$v_{\max}$	C	ρ
S01	`CSTR5`	RIC	3.06201	9.31e-7	C01	1.13†
S02	`CSTR5`	GLOA	3.13019	9.99e-7	C01	2.23e+3†
S03	`CSTR5`	LOA	3.06201	9.31e-7	C01	1.13†
S05	`CSTR5`	Hull	3.06201	9.34e-7	C01	1.09†
S06	`CSTR10`	RIC	2.88954	7.87e-11	C02	4.30†
S07	`CSTR10`	GLOA	3.44794	9.16e-7	C02	1.93e+4†
S08	`CSTR10`	LOA	2.88954	7.87e-11	C02	4.30†
S10	`CSTR10`	Hull	2.88951	9.37e-7	C02	3.54†
S11	`CSTR15`	RIC	2.82939	1.39e-9	C03	1.30†
S12	`CSTR15`	GLOA	3.56190	9.78e-7	C03	2.59e+4†
S13	`CSTR15`	LOA	2.82939	1.39e-9	C03	1.30†
S16	`CSTR20`	RIC	2.79941	4.47e-9	C04	1.67†
S17	`CSTR20`	GLOA	3.61603	8.93e-7	C04	2.92e+4†
S18	`CSTR20`	LOA	2.79941	4.47e-9	C04	1.67†
S21	`CSTR25`	RIC	2.78148	9.72e-11	C05	1.72†
S22	`CSTR25`	GLOA	3.64775	9.83e-7	C05	3.11e+4†
S23	`CSTR25`	LOA	2.78148	9.72e-11	C05	1.72†
S27	`CSTR30`	GLOA	3.66862	9.40e-7	C06	3.25e+4†
S45	`Dynamic`	Hull	-196.931	1.00e-7	C09	7.11e+3†

Table S8: Saved feasible comparison solutions. The process `SCIP` analysis linked in Section S8 identifies each full vector and its original-GDP recheck. Displayed objectives and violations are rounded.

ID	Model	Method	Recorded f_w	$v_{\max}$
C01	`CSTR5`	Big-M	3.06198	9.40e-7
C02	`CSTR10`	Big-M	2.88941	9.76e-7
C03	`CSTR15`	Hull	2.82936	9.47e-7
C04	`CSTR20`	Hull	2.79936	9.90e-7
C05	`CSTR25`	Hull	2.78143	9.90e-7
C06	`CSTR30`	Big-M	2.76957	9.68e-9
C07	`Column`	GLOA	22271.5	1.00e-8
C08	`Small Batch`	GLOA	167428	9.00e-7
C09	`Dynamic`	Big-M	-212.009	1.00e-7

Additional tightened-feasibility check for `CSTR15`. A separate Dantzig `IPOPT` 3.14.19 solve with `MUMPS` 5.8.1 retained the archived LOA witness W03's discrete configuration and continuous starting values and targeted an original-GDP residual of 10^{-8} (SECQUOIA, 2026b). That historical witness is distinct from the

current SCIP comparison solution C03. Only exactly satisfied constant rows were omitted from the computational clone, preserving the fixed-configuration feasible set and all original constraints in the independent check. The solve returned objective 2.82939. The original-GDP recheck gave maximum unscaled violation 1.11×10^{-16}, passing the 10^{-8} target.

The current SCIP-based GLOA lower bound 3.56190 exceeds this tightened objective by 0.732507, giving $\rho \approx 2.59 \times 10^4$. This comparison preserves the earlier validation result and does not establish exact feasibility or the cause of the bound discrepancy.

Additional formulation check for Dynamic Hull. The saved Big-M point has objective -212.009, approximately 15.0776 below the Hull lower bound -196.931. It satisfies both the original GDP and a reconstructed Hull formulation at 10^{-5}, with maximum violation approximately 10^{-7} after assigning the auxiliary variables from their definitions. The reconstructed export matches the archived preflight export, but the compiled model from the actual solve was not retained. The check therefore confirms the numerical inconsistency without identifying its solver or interface cause; no optimization was rerun.

S10 Additional candidate-search and sampling comparisons

S10.1 Paired differences and sampling controls

Paired success differences retain the common case, budget and seed. For Grid, the descriptive percentile bootstrap resamples the 21 intact function vectors, each containing ten paired seed differences, rather than treating 210 observations as exchangeable Bernoulli draws. For Process, paired seeds are resampled within each of the 14 conditions; the six related Dynamic budgets are not pooled for inference. The seed is 20261020 with 2,000 bootstrap replications and deterministic case/seed ordering. Intervals are exploratory and unadjusted for multiple comparisons; ten-seed boundary cases may yield degenerate intervals, and zero inclusion does not establish equivalence. Current Grid and D-SDA process paired comparisons are provided by the publication analysis linked in Section S8.

Table S9: Selected paired success differences, in percentage points. The first method minus the second defines the sign; Process intervals concern the stated condition only.

Grid

B	Contrast	Difference	95% interval
10	RBFOpt – Uniform	13.33	[5.71, 24.29]
100	RBFOpt – Uniform	60.00	[42.86, 75.71]
10	BAM – BAM-prefix control	0.00	[0.00, 0.00]
100	BAM – BAM-prefix control	10.95	[0.48, 23.81]
10	FMQA-SA – Uniform	0.95	[-0.95, 2.86]
100	FMQA-SA – Uniform	40.00	[28.57, 52.38]
10	D-SDA – RBFOpt	-13.81	[-24.29, -6.19]
100	D-SDA – RBFOpt	-33.33	[-48.10, -19.52]

Process: ten paired seeds per condition

Case	B	Contrast	Outcomes	Difference	95% interval
Dynamic	12	RBFOpt – D-SDA	4/6/0/0	60	[30, 90]
Small Batch	10	FMQA-SA – Uniform	2/4/2/2	20	[-30, 60]

Outcomes list both successful / first only / second only / neither successful. The descriptive intervals resample functions for Grid and paired seeds within each Process condition; they do not establish general superiority or equivalence.

S10.2 Grid suite exclusions

Excluding Xin-She Yang 3 and Bohachevsky 2 preserves the main descriptive pattern. For the sampled Bohachevsky 2 lattice, $\cos(3\pi a_i) = 1$ at every coordinate, leaving an objective nearly redundant with sampled

Sum of Squares after normalization. Without both functions, $B = 100$ rates are 79.5% for `RBFOpt`, 66.8% for `BAM`, 57.9% for `FMQA-SA`, 54.7% for the `BAM`-prefix control, 42.6% for `D-SDA`, and 17.9% for uniform. At $B = 10$, `BAM` and its prefix control each retain 47.9%, compared with `RBFOpt` 15.8%.

S10.3 Limited reconstructed FMQA diagnostics

Seven FM models were reconstructed locally from archived observations in three selected runs because the original fitted weights and QUBOs were not saved (SECQUOIA, 2026c). All 3,328 enumerated binary states passed the recorded energy-conversion checks. Five selected SA proposals minimized their reconstructed models over eligible candidates; two `Dynamic` fallback iterations had no eligible stored SA sample. These selected checks do not verify the historical weights or isolate the contributions of learning, annealing, and fallback. Synthetic zero-coefficient tests produced nonfinite scaling and a wrapper error before annealing or fallback; whether this occurred in the historical fits is unknown.